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Effects and neural mechanisms of a brain–computer interface-controlled soft robotic glove on upper limb function in patients with subacute stroke: a randomized controlled fNIRS study

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Abstract

Background and purpose The brain-computer interface-based soft robotic glove (BCI-SRG) holds promise for upper limb rehabilitation in subacute stroke patients, yet its efficacy and neural mechanisms are unclear. This study aimed to investigate the therapeutic effects and neural mechanisms of BCI-SRGs by functional near-infrared spectroscopy (fNIRS).

Methods Forty subacute stroke patients with left-sided hemiparesis were randomized into the BCI-SRG (n = 20) and soft robotic glove (SRG) (n = 20) groups. Both groups received 20 sessions of intervention over 4 weeks in addition to conventional rehabilitation. The BCI-SRG group was trained using a soft robotic glove controlled by a brain–computer interface (BCI), whereas the SRG group used the same soft robotic glove without BCI control. The clinical outcomes included the Action Research Arm Test (ARAT), the Fugl-Meyer Assessment Upper Limb (FMA-UL), and Modified Barthel Index (MBI) scores. In addition, fNIRS was used to explore potential clinical brain mechanisms. All assessments were performed before treatment and after 4 weeks of treatment.

Results A total of 39 participants completed the intervention and clinical assessments (BCI-SRG: n = 20; SRG: n = 19). Compared with the SRG group, the BCI-SRG group showed greater improvements in the ARAT ($Z = -2.139$, $P = 0.032$) and FMA-UL ($Z = -2.588$, $P = 0.010$), with no notable difference in the MBI ($Z = -1.843$, $P = 0.065$). fNIRS data were available for 35 participants (BCI-SRG: n = 17; SRG: n = 18). Within-group comparisons revealed significant postintervention increases in cortical activation in the bilateral sensorimotor cortex (SMC) and medial prefrontal cortex (MPFC) in the BCI-SRG group, whereas no significant changes were observed in the SRG group. Between-group comparisons

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further revealed significantly greater changes in HbO concentrations in the BCI-SRG group than in the SRG group across the same cortical regions. Moreover, changes in prefrontal activation (post–pre) were positively correlated with improvements in ARAT scores, with significant correlations observed in the left dorsal lateral prefrontal cortex (LDLPFC) (Ch9, r = 0.592, $P = 0.012$; Ch25, r = 0.488, $P = 0.047$) and right dorsal lateral prefrontal cortex (RDLPFC) (Ch19, r = 0.671, $P = 0.003$).

Conclusions BCI-SRG training significantly enhances upper limb function and facilitates bilateral motor and sensory cortical reorganization. PFC activation is correlated with functional improvements, suggesting a potential mechanism underlying the benefits of rehabilitation in stroke patients.

Trial registration This trial was registered under the Chinese Clinical Trial Registry (ChiCTR2400082786) and was retrospectively registered on April 8, 2024.

Keywords Brain–computer interface, Soft robotic glove, Stroke, Upper limb, Functional near-infrared spectroscopy

Introduction

Stroke is one of the leading causes of disability globally [1]. Restoring motor function, particularly in patients with affected upper limbs, is critical for improving quality of life and enabling patients to regain independence [2]. Clinical guidelines recommend that stroke should engage in training that is task-specific, repetitive, and actively engaging to enhance upper limb sensorimotor function [3]. Among these principles, active participation is particularly critical and has been shown to be essential for driving cortical reorganization and facilitating functional recovery [4]. However, motor impairments after stroke may limit patients'ability to engage voluntarily in therapy. To address this challenge, brain-computer interface (BCI) systems have gained increasing attention for their potential to translate patient intention into motor output and foster active engagement in rehabilitation.

Among the various BCI modalities, motor imagery-based systems are commonly used approaches in post-stroke rehabilitation. These systems capture electroencephalography (EEG) signals during motor imagery, which are then decoded and analyzed to assess the accuracy of the patient's motor intention during imagined movements [5]. They are often integrated with external devices, such as exoskeletons, and provide real-time feedback that enhances rehabilitation and motor function [6]. However, these exoskeletons have limitations, including bulkiness and restricted range of motion, which hinder the patient's ability to perform fine motor tasks. Soft robotic hands, in contrast, offer an adaptable alternative, with greater dexterity and the ability to closely mimic natural hand movements, particularly for the training of fine motor skills in the hands [7]. This makes brain–computer interface-based soft robotic glove (BCI-SRG) a desirable intervention. Some studies have explored the use of BCI-SRGs for upper limb rehabilitation in stroke patients. Cheng et al. [8] conducted a randomized controlled trial with 11 chronic stroke patients and demonstrated significant improvement in upper limb motor function in the BCI-SRG group compared with the SRG group. Similarly, Guo et al. [9] reported that the BCI-SRG led to better upper limb function recovery than the SRG in 30 chronic stroke patients. Despite these promising results, studies are limited by small sample sizes and a focus on chronic stroke patients. The efficacy of the BCI-SRG in patients with subacute stroke remains to be explored, and the mechanisms underlying its effectiveness are still unclear.

Motor recovery after stroke is closely related to neuroplastic changes in the adult brain, with specific cortical activation playing a key role in recovery [10]. While several studies have demonstrated the clinical efficacy of BCI interventions in enhancing motor recovery post-stroke [11, 12], the underlying neural mechanisms remain insufficiently understood. Neuroimaging techniques such as functional magnetic resonance imaging (fMRI) and functional near-infrared spectroscopy (fNIRS) are valuable tools for investigating these mechanisms [13, 14]. Compared with fMRI, fNIRS offers several advantages. It is portable, non-invasive, and can be used in a variety of settings, including standing or during active movement, without the need for physical restraint. This flexibility is especially useful in stroke rehabilitation, where patients may experience mobility limitations. Moreover, fNIRS provides higher temporal resolution (~ 10 Hz) than does fMRI, allowing for the continuous tracking of brain activity during task execution [15]. Studies have demonstrated that fNIRS can successfully assess functional connectivity during both the resting and active states, revealing how different brain regions synchronize during motor tasks [16]. These features make fNIRS an effective and practical tool for understanding brain function and recovery after stroke, facilitating personalized rehabilitation strategies.

This prospective, single-centre, randomized controlled clinical trial (RCT) aims to compare the efficacy of BCI-SRG and SRG interventions for facilitating upper limb functional rehabilitation in stroke patients. In addition to clinical functional performance, fNIRS was effectively

employed to detect haemoglobin changes in patients subjected to BCI-SRG and SRG therapy in the present study. We hypothesized that the BCI-SRG would lead to greater improvements in upper limb function and outperform the SRG alone by remodelling motor-related brain areas.

Methods

Participants

This two-arm, parallel RCT was conducted in compliance with the Declaration of Helsinki and registered with the Chinese Clinical Trial Registry (ChiCTR2400082786). The study received approval from the Ethics Committee of Wuxi Central Rehabilitation Hospital (WXMHCIRB2024LLky018) and adhered to the CONSORT reporting guidelines (Fig. 1, Supplementary Table S1). A total of 40 stroke patients were enrolled from the Rehabilitation Medicine Department of Wuxi Central Rehabilitation Hospital, China, between April and December 2024. Table 2 presents the demographic and clinical characteristics of the participants.

The inclusion criteria were as follows: (1) unilateral subcortical lesions in the right hemisphere confirmed by MRI or computed tomography; (2) aged between 30 and 80 years; (3) stroke onset within 2 weeks to 3 months; (4) persistent hemiplegia causing hand dysfunction, with Brunnstrom hand stages II-IV, and at least one of the following motor abilities: (a) partial upper limb function [Action Research Arm Test (ARAT) gross motor score $\geq$ 3]; (b) manual muscle test (MMT) score $\geq$ 2 on shoulder flexion and either elbow flexion or extension; or (c) active shoulder flexion or abduction through $\geq$ 50% of the passive range of motion in gravity-eliminated positions, combined with any of the following motions: elbow flexion, elbow extension, wrist flexion, wrist extension, finger flexion, or finger extension [17]. (5) Passive range of motion in the affected upper limb near normal; (6) Mini-Mental State Evaluation (MMSE) score > 20 [18], with the ability to understand and cooperate with assessments and training; (7) motor imagery ability assessed using the Kinesthetic and Visual Imagery Questionnaire (KVIQ-20), with a score of > 55 indicating sufficient proficiency [19]. (9) Written informed consent was obtained from the participants or their guardians.

The exclusion criteria included the following: (1) coexisting neurological disorders such as peripheral nerve injuries, Parkinson's disease, or multiple sclerosis; (2) severe dysfunction or failure of major organs (heart, liver, kidney, etc.); (3) cognitive impairments, aphasia, or conditions preventing understanding or following instructions; (4) posterior circulation infarction; (5) severe upper limb spasticity (modified Ashworth $\geq$ 3) or joint contractures; (6) skin injuries, infections, or hypersensitivity to pain; and (7) pregnancy.

Sample size calculation

Due to the lack of prior reports on the efficacy of brain-computer interface-based soft robotic gloves for upper limb function in subacute stroke patients, it was impractical to determine an exact sample size a priori. Therefore, we set 18 participants per group (36 total), which aligns with the minimum recommended sample size for pilot trials [20]. Accounting for a potential 10% dropout rate, we plan to enroll 40 participants in this pilot study.

Experimental procedure

The participants were randomly assigned to either the BCI-SRG group or the soft robotic glove (SRG) group by one of the authors (L.X. or Y.X.), who was not involved in the intervention or evaluation and used a computer-generated, sealed-envelope method without adjustment factors. Figure 1 shows a schematic overview of the experimental procedure. Both groups received conventional upper-limb rehabilitation, including therapist-guided active and passive limb exercises, task-oriented exercises, ADL training, and neuromuscular electrical stimulation. Therapy was delivered for 60 min per day, 5 days per week. In addition to traditional rehabilitation, the BCI-SRG group received training with a soft robotic glove by a BCI, whereas the SRG group used the same glove operated by a soft robotic control system. Both groups underwent 20 intervention sessions over 4 weeks (5 sessions per week), with motor function assessments conducted at two time points: preintervention and 4 weeks postintervention. fNIRS was used to assess brain activity at baseline and after 4 weeks.

BCI-SRG training

A 20-channel active electrode system (Neuracle Technology Co., Ltd., China) was used to acquire continuous EEG signals. During the acquisition, twenty channels (Fp1, Fp2, F7, F3, Fz, F4, F8, T3, C3, Cz, C4, T4, T5, P3, Pz, P4, T6, O1, Oz, and O2 (Fig. 2b)) were selected, the sampling rate was 1000 Hz, and the electrode impedance was maintained below 10 kΩ to ensure signal quality. The signals were bandpassed with a 2–60 Hz filter and a notch filter (48–52 Hz) to remove artefacts and power line interference, respectively. The soft robotic glove module (Nanjing Reseader Medical Technology Co., Ltd., China) (Fig. 2c), comprising the robotic glove control box and the soft robotic glove, was responsible for providing mechanical feedback to the participants.

To enable training-free online motor imagery classification, we initially employed mu suppression scores derived from the C3 and C4 channels to quantify the event-related desynchronization magnitude and established a threshold for online classification.

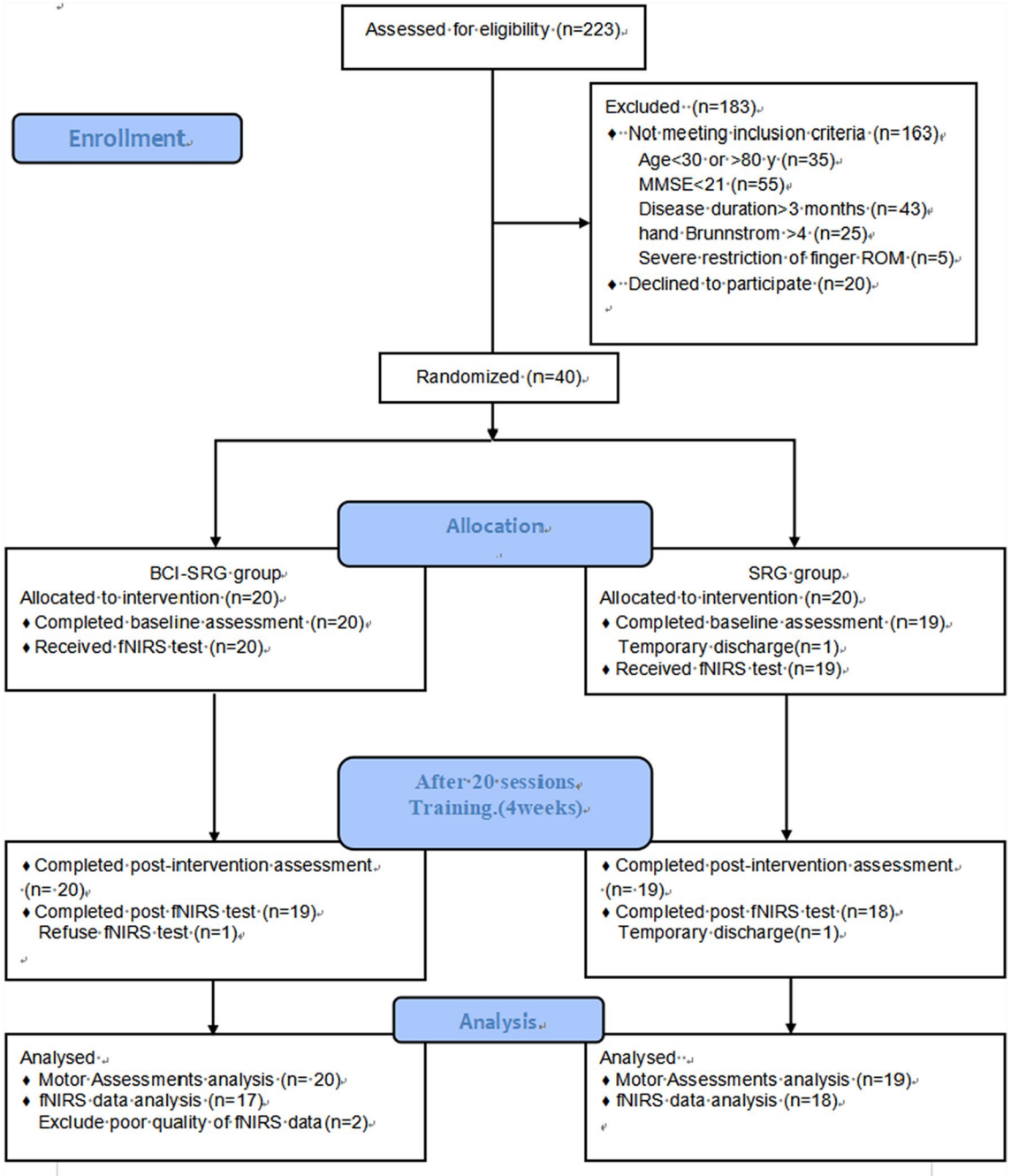


Fig. 1 CONSORT flow diagram

The target channel data were transformed into the frequency domain using the Fast Fourier Transform, which enabled the calculation of energy within the mu band. Mu suppression was computed using the formula: $\text{Musupp} = -(\text{Mup}_{\text{task}} - \text{Mup}_{\text{rest}})/\text{Mup}_{\text{rest}}*100\%$, where Mup_{task} represents the EEG energy during the MI task and where Mup_{rest} denotes the EEG energy during the resting state.

Before training, the therapist provides detailed instructions on the treatment procedure. Following this, the patient is asked to focus on a black cross displayed on the screen for 60 s to collect resting-state EEG data. After 2 s of preparation, each session begins with an image of a cup displayed on the screen, prompting the patient to imagine grasping the cup with their affected hand. This task lasts for 5 s. If successful, a 15-s congratulatory

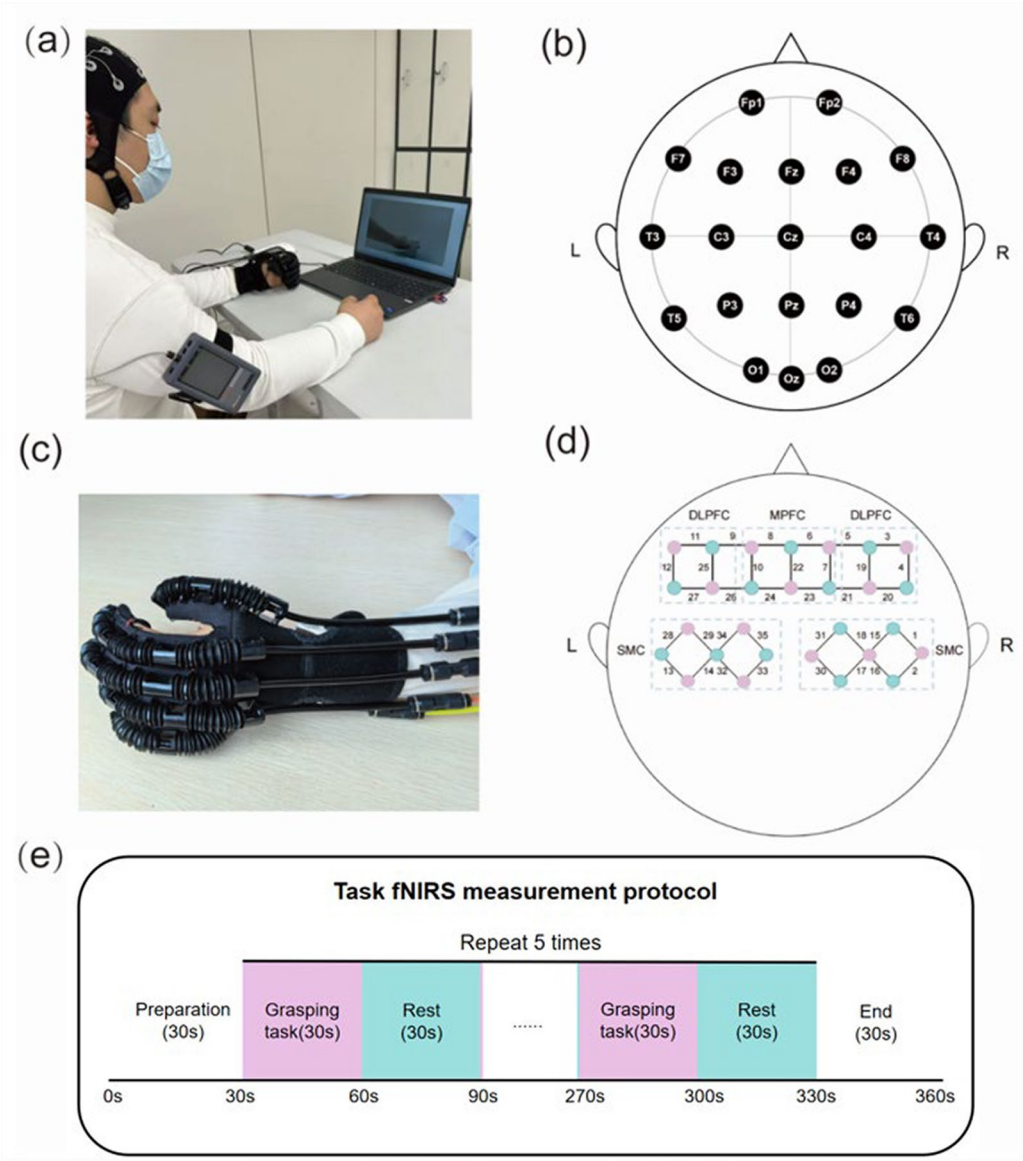


Fig. 2 System design and protocol for BCI-SRG training. **a** A participant engaged in the BCI-SRG training session and wore the soft robotic glove while EEG signals were recorded. **b** EEG electrode placement according to the 10–20 system. **c** Soft robotic glove worn by the participant and equipped with sensors for feedback. **d** Configuration of fNIRS channels. **e** Task fNIRS measurement protocol, detailing a 5-cycle procedure of preparation, grasping task, rest, and end. L: left hemisphere; R: right hemisphere

animation is shown, and soft robotic hand assistance is triggered to aid the patient in performing the task. If unsuccessful, a 2-s encouraging animation is provided. After the animation, the patient rested for 5 s before the next trial. Each training session consisted of 75 trials, which were grouped into 15 rounds of 5 trials each. The average session duration in the BCI-SRG group was approximately 20–25 min, depending on the proportion of successful detections. The detailed procedure is illustrated in Fig. 3.

SRG training

In the SRG training group, patients were seated comfortably in front of a screen wearing a soft robotic glove. Training consisted of 10-s gripping tasks followed by 5-s rest intervals, with a total of 75 repetitions, lasting 20 min in total. A left-hand grasping video was displayed on the screen, and patients were instructed to imagine performing the grasping motion with their left hand in synchronization with the video.

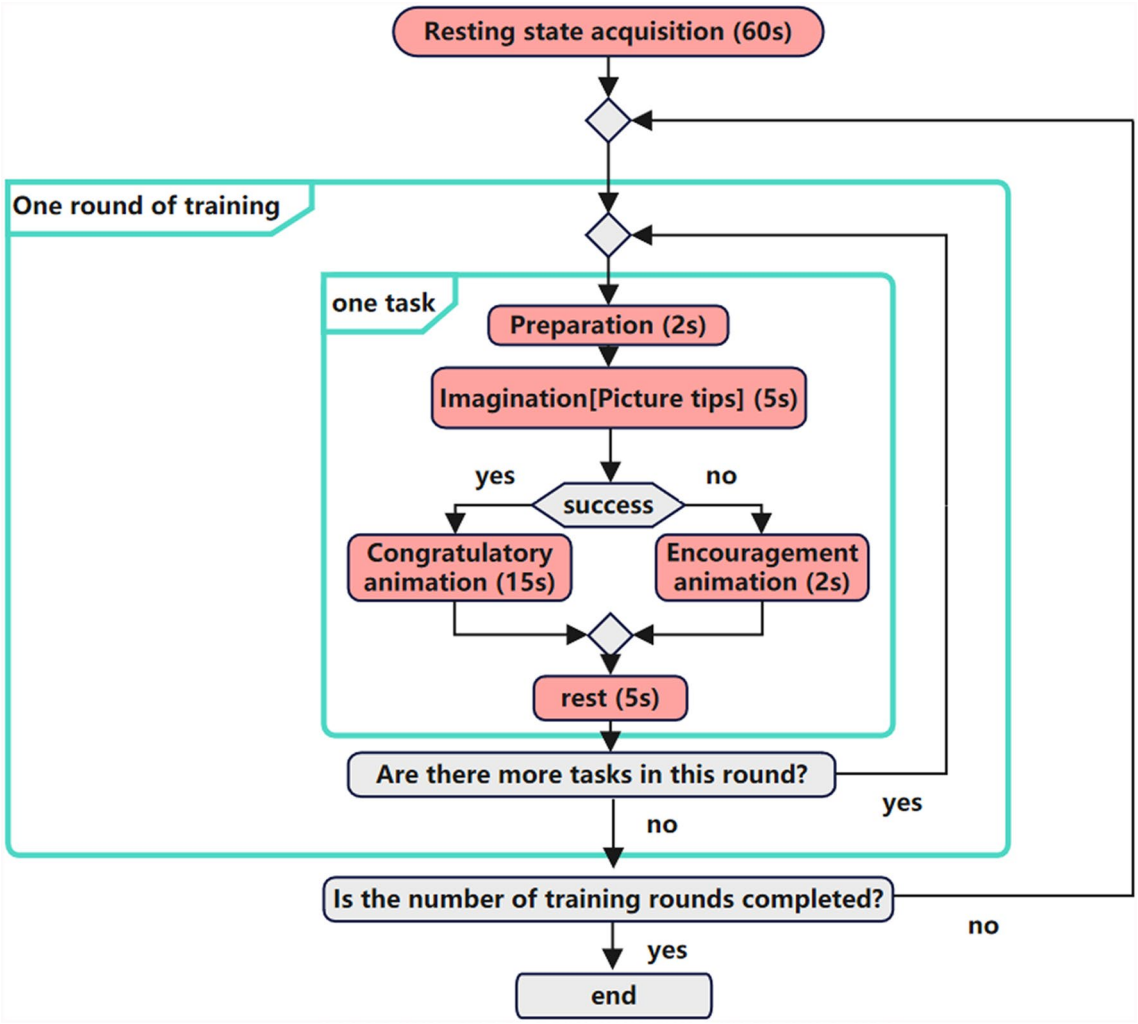


Fig. 3 Flowchart of the BCI-SRG training protocol, outlining the sequence of tasks: preparation, motor imagery, task execution, and feedback

Clinical assessments

The primary outcome of this study was the ARAT score, a validated measurement of upper limb motor function with proven high reliability and validity in stroke patients [21]. The ARAT consists of 19 items across four subscales: grasp, grip, pinch, and gross movements. Each item is scored on a 4-point scale (0–3), with a maximum score of 57 for one upper limb. Higher scores indicate better function. Even if some upper limb motor function is present, a score of zero is assigned in this assessment if the hand is unable to perform grasping, gripping, or pinching tasks. The secondary outcomes included the FMA-UL and MBI scores. Additionally, cortical activation changes were evaluated using fNIRS to assess motor cortical activity at the first and final neurofeedback sessions.

fNIRS data acquisition

A continuous-wave fNIRS system (NirSmart, Danyang Huichuang Medical Equipment Co., Ltd., Zhenjiang, China) with wavelengths of 730 and 850 nm was used to measure changes in the oxygenated haemoglobin (Δoxy-Hb) concentration, with a sampling rate of 11 Hz. fNIRS measurements were conducted by trained researchers (W.Z.). A total of 35 channels were set up, including 14 light sources and 14 detectors. The distance between each detector and the light source was 30 mm. On the basis of the standard Brodmann brain mapping method, the channels were divided into seven regions of interest (ROIs): the LDLPFC (9, 11, 12, 25, 26, 27), MPFC (6, 7, 8, 10, 22, 23, 24), RDLPFC (3, 4, 5, 19, 20, 21), LSMC (13, 14, 28, 29, 32, 33, 34, 35), and RSMC (1, 2, 15, 16, 17, 18, 30, 31). The montage

Table 1 NIRS probe channel assignment to regions of interest

ROI	Number	Channels
LDLPFC	6	CH9/11/12/25/26/27
MPFC	7	CH6/7/8/10/22/23/24
RDLPFC	6	CH3/4/5/19/20/21
LSMC	8	CH13/14/28/29/32/33/34/35
RSMC	8	CH1/2/15/16/17/18/30/31

LDLPFC, left dorsal lateral prefrontal cortex; MPFC, medial prefrontal cortex; RDLPFC, right dorsal lateral prefrontal cortex; LSMC, left sensorimotor cortex; RSMC, right sensorimotor cortex

configuration for the NIRS probe channels is shown in Fig. 2d and Table 1. Prior to the experiment, all the participants were instructed to sit still in a noise-free environment for 10–20 min to minimize haemodynamic responses caused by physical activity. A NIR gain quality check was conducted before each recording session to ensure optimal data acquisition, preventing both insufficient and excessive gain. To position the probes accurately on the scalp, the participants' heads were covered with a cap and secured using straps to adjust and stabilize the emitters and receivers. The hair was cleared to maximize light–tissue coupling and ensure close contact between the probes and the skin. fNIRS testing consisted of a baseline phase (30 s), a task phase (300 s), and an end phase (30 s). The task phase included five blocks, each comprising a 30-s grasp task followed by a 30-s rest period, for a total of 300 s (Fig. 2e).

fNIRS data processing

Data inspection and preprocessing were performed using the Homer2 toolbox in MATLAB 2014a (The MathWorks Inc.), with the following steps: (1) Channel quality check: All channels from each participant were inspected for quality, and the results were recorded before preprocessing. (2) Signal conversion: Raw NIRS intensity values were converted to optical density signals. (3) Motion artefact removal: Motion artefacts induced by head movement during data acquisition were detected and corrected using the motion artefact removal algorithm with the following parameters: tMotion = 0.5, tMask = 2.0, STDEVthresh = 20.0, AMPthresh = 0.5, and pSpline = 0.99. (4) Spline interpolation: Following motion artefact detection, spline interpolation was applied to correct for any remaining artefacts. The detection window was then shifted to the next segment until the entire time series was processed. (5) Filtering: A bandpass filter (0.01–0.1 Hz) was applied to remove physiological noise and drift, including heart rate (~ 1 Hz), respiration (0.2–0.3 Hz), and Mayer waves (~ 0.1 Hz) [22]. (6) Conversion to HbO concentration: Applying the modified Beer–Lambert law, the filtered optical density data were converted to HbO concentrations [23].

Statistical analysis

Data analysis was performed using SPSS 22.0 (IBM SPSS Statistics, Chicago, IL, USA) for statistical analysis and GraphPad Prism 8 (GraphPad Software Inc., San Diego, CA, USA) for visualization. The normality of the data distribution was assessed using the Shapiro–Wilk test. Demographic data are presented as the means ± standard deviations (continuous variables were analysed via independent-sample *t* tests) and counts (categorical variables were analysed by the chi-square test). Nonnormally distributed data (ARAT, FMA-UL, and MBI) are expressed as the median results and quartiles. The Wilcoxon rank sum test was used for intragroup comparisons, and the Mann–Whitney U test was used for intergroup comparisons.

fNIRS analysis was performed with MATLAB. HbO changes during the grasping task were analysed by extracting signals from each channel on the basis of predefined markers. The mean HbO values across five grasping trials were calculated. Intragroup contrasts (post–pre) were assessed using paired *t* tests, whereas between-group contrasts of change scores (post minus pre) were analysed using independent *t* tests. Multiple comparisons were corrected by the Benjamini–Hochberg procedure to control the false discovery rate (FDR), with significance set at $P < 0.05$. For visualization of brain activation, t statistic maps were generated using BrainNet Viewer. Significant t values were retained, whereas nonsignificant values were set to zero. The adjusted t values were mapped onto brain models, with colour gradients reflecting the magnitude of activation changes. Spearman correlation analyses were performed to examine the relationships between changes in HbO levels (post–pre) during grasping tasks and improvements in the ARAT score within each group. Statistical significance was set at $P < 0.05$.

Results

Demographics

A total of 40 eligible patients met the inclusion and exclusion criteria and participated in the study. However, during the intervention, one participant (from the SRG group due to exacerbation) withdrew from the trial. The overall attrition rate was 2.5%. As a result, a total of 39 participants were identified in this study (Fig. 1). Demographic characteristics are presented, and the homogeneity of the groups at baseline is confirmed in Table 2. The

Table 2 Demographic and clinical characteristics

Characteristics	BCI-SRG (n = 20)	SRG (n = 19)	t/x²	P
Age, years	61.75 ± 10.35	60.05 ± 14.35	0.175	0.862[a]
Sex, n (%)			1.642	0.200[b]
Male	12 (60.0)	15 (78.9)		
Female	8 (40.0)	4 (21.1)		
Stroke type, n (%)			0.641	0.423[b]
Ischaemic	15 (75.0)	12 (63.2)		
Haemorrhagic	5 (25.0)	7 (36.8)		
Time from the onset, days	49.50 ± 23.36	42.42 ± 24.50	0.924	0.362[a]
BMI (kg/m²)	24.32 ± 2.66	24.61 ± 2.51	−0.355	0.725[a]
MMSE	24.40 ± 2.70	24.63 ± 2.93	−0.257	0.799[a]
ARAT	3.00 (3.00, 6.00)	5.00 (3.00, 7.00)	−0.898	0.369[c]
MBI	33.00 (30.25, 35.00)	33.00 (30.00, 37.00)	−0.283	0.777[c]
FMA-UL	22.00 (18.50, 27.00)	21.00 (19.00, 25.00)	−0.339	0.734[c]

[a] independent-sample *t* test; [b]Pearson's chi-square test; [c]Mann–Whitney U test

Table 3 Results of the clinical scores

Characteristics		BCI-SRG (n = 20)	SRG (n = 19)	Z	P
ARAT	T1	3.00(3.00, 6.00)	5.00(3.00, 7.00)	−0.898	0.369[a]
	T2	17.00(10.00, 34.75)	11.00(6.00, 21.00)	−1.689	0.091[a]
	change	14.00(7.00, 28.50)	7.00(3.00, 17.00)	−2.139	0.032[a]
	Z	−3.727	−3.624		
	P	< 0.001[b]	< 0.001[b]		
FMA-UL	T1	22.00(18.50, 27.00)	21.00(19.00, 25.00)	−0.339	0.734[a]
	T2	39.50(29.75, 54.50)	31.00(24.00, 45.00)	−2.109	0.035[a]
	change	18.00(11.00, 25.00)	7.00(5.00, 18.00)	−2.588	0.010[a]
	Z	−3.923	−3.826		
	P	< 0.001[b]	< 0.001[b]		
MBI	T1	43.00(31.50, 51.25)	38.00(35.00, 50.00)	−0.169	0.866[a]
	T2	64.00(57.00, 70.00)	57.00(52.00, 65.00)	−1.477	0.140[a]
	change	22.00(17.25, 28.50)	19.00(10.00, 26.00)	−1.843	0.065[a]
	Z	−3.922	−3.824		
	P	< 0.001[b]	< 0.001[b]		

BCI, Brain-computer interface; SRG, soft robotic glove; ARAT, action research arm test; FMA-UL, Fugl-Meyer assessment for upper limb; MBI: modified Barthel index. T1: preintervention; T2: postintervention; change value: difference between T2 and T1. [a]Mann–Whitney U test, [b]Wilcoxon rank sum test

treatment protocol was well tolerated, and no device-related malfunctions, safety concerns, or other adverse events were reported in either group.

Clinical outcome measures

Baseline comparisons revealed no significant differences between the groups, confirming their statistical homogeneity (Table 3). Clinical changes after training were observed, with increased ARAT, FMA-UL, and MBI scores in both groups, indicating improved upper limb motor function and ability to perform activities of daily living. In the posttreatment assessment, although between-group comparisons of the ARAT and MBI scores revealed no significant differences, the BCI-SRG group presented significantly higher FMA-UL scores than did the SRG group ($Z = -2.109$, $P = 0.035$, Table 3). Moreover, the improvement in the ARAT score ($Z = -2.139$, $P = 0.032$, Fig. 4a) and FMA-UL score ($Z = -2.588$, $P = 0.010$, Fig. 4b) from baseline to postintervention was significantly greater in the BCI-SRG group than in the control group, whereas no significant difference in the MBI score ($Z = -1.843$, $P = 0.065$, Fig. 4c) was detected.

Cortical activation

Owing to poor data quality in two patients and one refusal of fNIRS assessment in the BCI-SRG group, along with one temporary discharge in the SRG group, a total of 17 participants in the BCI-SRG group and 18 in the SRG group were included in the final analysis.

Within-group comparisons revealed no significant cortical activation in any channels in the SRG following the intervention ($P > 0.05$). In contrast, the BCI-SRG group demonstrated significant increases in cortical activation postintervention, with changes observed in nine channels localized to the bilateral sensorimotor cortex (SMC) and medial prefrontal cortex (MPFC)

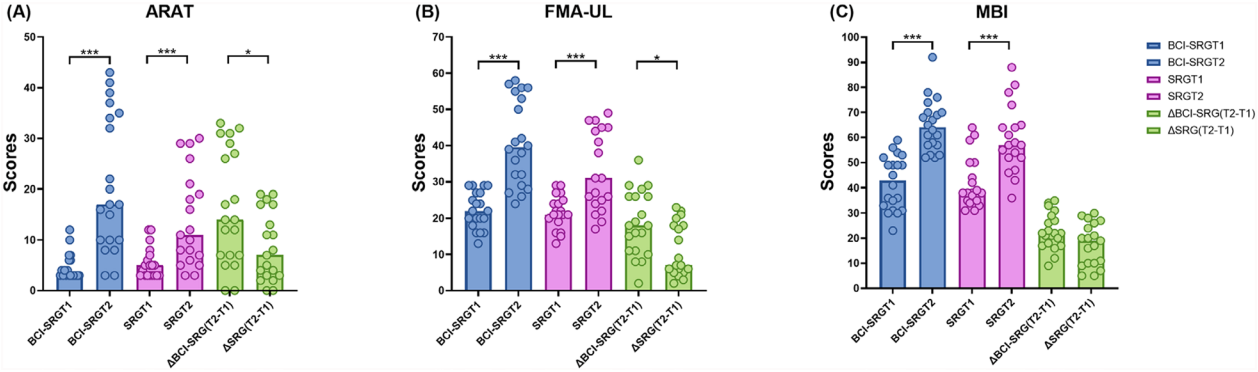


Fig. 4 Comparison of clinical scale scores between the BCI-SRG and SRG groups at different time points. **a** ARAT, action research arm test; **b** FMA-UL, Fugl-Meyer assessment for upper limb; **c** MBI, modified Barthel index. ***$P < 0.001$, *$P < 0.05$

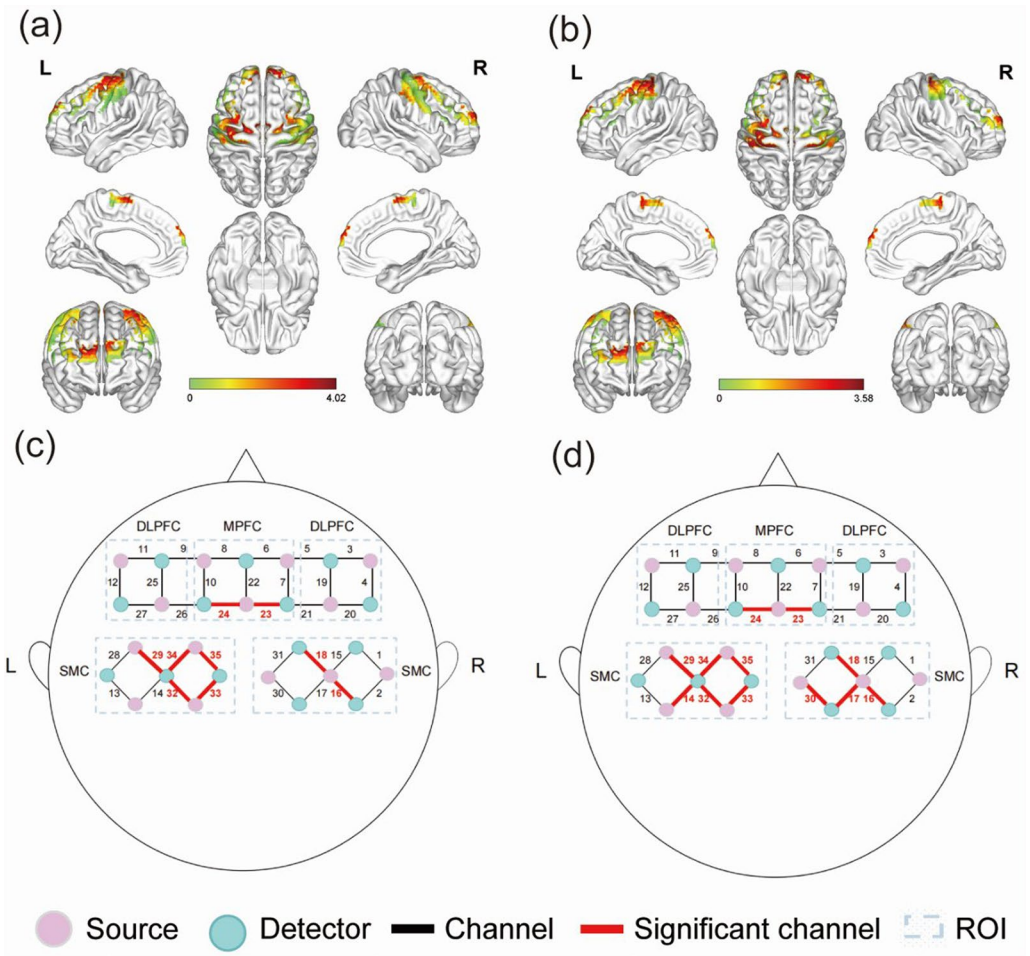


Fig. 5 Cortical activation maps of HbO levels. **a** Changes in brain activation in the BCI-SRG group before and after the intervention; brilliant colours (red to bright yellow) indicate increased cortical activation. **b** Group comparison of mean HbO activation differences during the grasping task, highlighting significantly higher activation levels in the BCI-SRG group than in the SRG group. The colour bar represents t values. **c** The channels with significant differences within the BCI-SRG group are highlighted in red. **d** Channels with significant differences between groups (BCI-SRG vs. SRG) are highlighted in red. L: left hemisphere; R: right hemisphere

(Fig. 5a). Specifically, these significantly activated channels were channel 16 (RSMC, t = 3.068, $P = 0.037$), channel 18 (RSMC, t = 4.024, $P = 0.024$), channel 23 (MPFC, t = 3.810, $P = 0.024$), channel 24 (MPFC, t = 2.888, $P = 0.042$), channel 29 (LSMC, t = 3.065, $P = 0.037$), channel 32 (LSMC, t = 3.427, $P = 0.024$), channel 33 (LSMC, t = 2.886, $P = 0.042$), channel 34 (LSMC, t = 3.452, $P = 0.024$), and channel 35 (LSMC, t = 3.591, $P = 0.024$) (Fig. 5c).

Between-group comparisons of the mean changes in HbO levels (post–pre) during gripping tasks revealed significantly greater alterations in 12 channels in the BCI-SRG group than in the SRG group (Fig. 5b). These between-group differences were primarily distributed across the bilateral SMC and MPFC regions, including channel 14 (LSMC, t = 2.858, $P = 0.033$), channel 16 (RSMC, t = 2.770, $P = 0.033$), channel 17 (RSMC, t = 2.719, $P = 0.033$), channel 18 (RSMC, t = 2.977, $P = 0.032$), channel 23 (MPFC, t = 3.248, $P = 0.031$), channel 24 (MPFC, t = 2.664, $P = 0.035$), channel 29 (LSMC, t = 2.729, $P = 0.033$), channel 30 (RSMC, t = 2.771, $P = 0.033$), channel 32 (RSMC, t = 3.538, $P = 0.022$), channel 33 (RSMC, t = 3.061, $P = 0.032$), channel 34 (RSMC, t = 3.576, $P = 0.022$), and channel 35 (RSMC, t = 3.018, $P = 0.032$) (Fig. 5d).

Correlation analysis of cortical activation changes and clinical scores

Spearman correlation analysis between the changes in HbO levels across channels during the grasping task and the ARAT score differences revealed no significant correlations in the SRG group. However, in the BCI-SRG group, significant correlations were found in three channels: the LDLPFC (Ch9, r = 0.592, p = 0.012; Ch25, r = 0.488, p = 0.047) and RDLPFC (Ch19, r = 0.671, p = 0.003) (Fig. 6).

Discussion

This study demonstrated the therapeutic potential and underlying neural mechanisms of a BCI-driven SRG system for poststroke upper limb rehabilitation, as assessed using fNIRS. Compared with the conventional SRG training group, the BCI-SRG group exhibited significantly greater improvements in upper limb function, as indicated by higher scores on the ARAT and FMA-UL. fNIRS data revealed robust cortical activation in the BCI-SRG group, particularly in the SMC and MPFC, whereas no comparable activation was observed in the conventional SRG group. Notably, activation in the bilateral prefrontal cortex (RDLPFC, LDLPFC) was positively correlated with ARAT scores, suggesting a relationship between prefrontal activity and upper limb functional recovery. These results underscore the promise of BCI-assisted rehabilitation in promoting functional gains through enhanced neural circuit reorganization and cognitive–motor integration.

Given the robust cortical activation observed in the BCI-SRG group and the absence of such activation in the SRG group, we further analysed within-group effects to characterize the distinct neural responses induced by each intervention. The results revealed that SRG training alone did not elicit significant cortical activation, whereas the BCI-SRG group exhibited significant activation across nine channels, including the bilateral SMC and MPFC. These findings suggest that motor imagery combined with the SRG, when delivered without neurofeedback, may be insufficient to engage cortical motor networks meaningfully. Although the SRG protocol required participants to visualize hand movements while observing a screen, the absence of real-time neurofeedback likely limited both sensory and cognitive engagement. In contrast, the BCI-SRG system establishes a closed-loop interaction by integrating both central (cortical activity) and peripheral (robotic movement) feedback, thereby

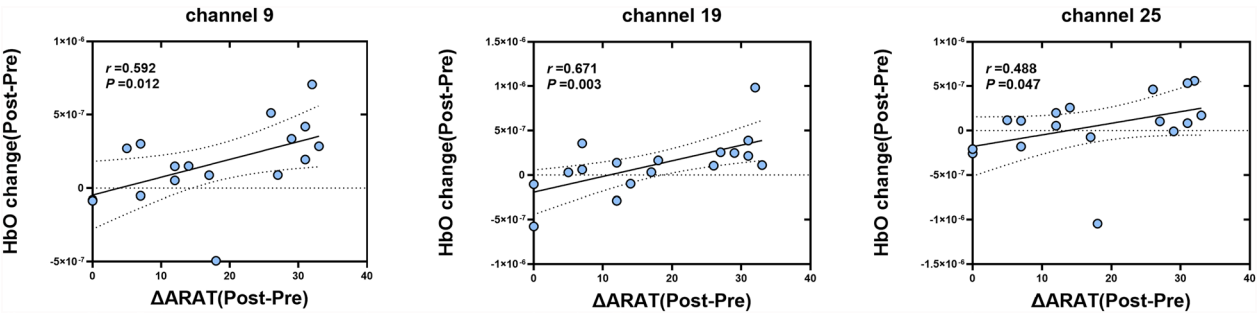


Fig. 6 Correlations between changes in HbO levels and ARAT score differences in the BCI-SRG group. Scatter plots show significant positive correlations between changes in HbO levels (Post–Pre) during the grasping task and ARAT score improvements (ΔARAT) in channel 9 (LDLPFC, r = 0.592, p = 0.012), channel 19 (RDLPFC, r = 0.671, p = 0.003), and channel 25 (LDLPFC, r = 0.488, p = 0.047). The solid lines represent the regression fit, and the dashed lines indicate the 95% confidence intervals. javascript:void(0);

reinforcing the link between motor intention and sensory consequences. This bidirectional feedback loop is critical for enhancing neuroplasticity, as it enables users to perceive the outcomes of their motor imagery in real time, thereby strengthening sensorimotor integration [24–26]. Studies have emphasized that motor learning is significantly influenced by error-based corrections and performance feedback [27, 28], both of which are lacking in purely imagination-based tasks that do not incorporate external feedback. Therefore, the key distinction between the BCI-SRG and SRG groups lies in the presence of real-time, bidirectional feedback, which appears to be essential for promoting cortical engagement and facilitating effective motor learning.

In this study, stroke patients who received BCI-driven motor training demonstrated notable improvements in upper limb function, accompanied by increased activation in the bilateral SMC and MPFC. The SMC, which encompasses the primary motor cortex and adjacent somatosensory areas, plays a central role in acilitating sensorimotor integration, improving fine motor skills, and facilitating motor learning [29–31]. Following a stroke, the SMC often undergoes substantial reorganization as part of the brain's intrinsic repair mechanisms [32]. Mounting evidence suggests that both the extent and pattern of SMC activation are tightly linked to motor recovery outcomes. For example, an animal model study demonstrated that poststroke motor deficits and recovery are closely associated with disruption and subsequent restoration of functional connectivity between the SMC and other brain regions [33]. In a human study using task-state fMRI, stroke patients who received combined mental and physical practice training exhibited enhanced activation in the contralateral SMC, with the extent of activation positively correlated with hand function improvement [34]. Furthermore, fNIRS research has shown that stroke survivors display extensive bilateral SMC activation during both unimanual and bimanual motor tasks [35]. In our study, we also observed activation in the contralateral SMC, highlighting its important role in supporting recovery. This finding aligns with evidence that, while the majority of corticospinal fibres cross at the medullary pyramids, a portion remains ipsilateral, enabling bilateral motor control [36]. The enhanced contralesional activation we observed may reflect compensatory mechanisms that support the recovery of complex motor functions such as grasping, which require more extensive neural recruitment [37].

Furthermore, the PFC has emerged as a critical region in the recovery of upper-limb function following stroke [38]. This finding was further supported by our findings, which revealed that between-group differences in activation were most pronounced in the MPFC, and correlations with functional improvement (ARAT scores) were observed in the LDLPFC and RDLPFC. Although these regions are anatomically distinct, both are part of the prefrontal cortex and contribute to higher-order motor control [11, 39, 40]. Current insights into the role of the PFC in motor function following stroke are derived primarily from functional neuroimaging studies. These investigations have consistently demonstrated heightened PFC activation during various motor-related tasks, including visuomotor grip tasks [41], simple finger tapping [42], and action-selection tasks [43]. Notably, motor imagery in stroke patients has also been shown to recruit the PFC [44] and to enhance excitatory coupling with the dorsal premotor cortex and supplementary motor area, suggesting compensatory engagement of cognitive networks in motor planning and preparation [45]. It has been proposed that the PFC may facilitate increased neuronal output from the executive motor network to spinal motor neurons via projections originating in premotor regions [46]. These findings collectively underscore that motor recovery after stroke involves not only sensorimotor restoration but also higher-order cognitive processes essential for functional relearning. Our results align with prior evidence and underscore the role of BCI systems in engaging prefrontal circuits to strengthen cognitive–motor integration [18].

Despite these valuable findings, several limitations warrant consideration. First, to ensure accurate fNIRS analysis without the need for image flipping, only patients with right hemispheric lesions were included. This may limit the generalizability of the findings to individuals with left-hemispheric strokes. Second, the absence of a follow-up period prevents assessment of the long-term retention and durability of the observed therapeutic effects. Additionally, while the BCI-SRG system has demonstrated strong efficacy, its technical complexity may pose challenges for routine clinical implementation, particularly in resource-limited settings. Another important limitation is the lack of significant improvement in the MBI scores, despite substantial gains in ARAT scores. This disconnect suggests that BCI-driven training may not adequately address activities of daily living (ADL) function. The absence of fine finger dexterity training in our study, which is essential for functional independence, may explain the limited ADL gains. Such training is linked to improved hand use in chronic stroke patients [47].

In conclusion, this study demonstrated that BCI-SRG intervention enhances motor recovery in stroke patients by promoting cortical activation, particularly in the primary somatosensory, motor, and prefrontal areas. The observed correlation between prefrontal activation and ARAT scores highlights the role of cognitive control in motor rehabilitation. These findings emphasize the

importance of integrating cognitive–motor pathways, reinforcing the potential of BCI–SRG systems to promote neural circuit reorganization and enhance poststroke rehabilitation outcomes.

Abbreviations

ARAT	Action Research Arm Test
BCI-SRG	Brain-Computer Interface-based Soft Robotic Glove
EEG	Electroencephalography
FDR	False Discovery Rate
FMA-UL	Fugl-Meyer Assessment Upper Limb
fMRI	Functional magnetic resonance imaging
fNIRS	Functional near-infrared spectroscopy
HbO	Oxygenated haemoglobin
LDLPFC	Left dorsal prefrontal cortex
LSMC	Left sensorimotor cortex
MI-BCI	Brain-Computer Interface-based Motor Imagery
MMSE	Mini-Mental State Evaluation
MPFC	Medial prefrontal cortex
Δoxy-Hb	Changes in oxygenated haemoglobin
RCT	Randomized controlled trial
RDLPFC	Right dorsolateral prefrontal cortex
RSMC	Right sensorimotor cortex
SMC	Sensorimotor cortex
SRG	Soft Robotic Glove

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12984-025-01704-x.

Additional file 1.

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Author contributions
XJ and YS conceptualized and designed the study. XJ, XL, and YX contributed to participant recruitment and clinical data collection. HY and CY were responsible for the methodology and data analysis. WZ supervised the fNIRS data acquisition and preprocessing. HW, CR, and YJ were responsible for reviewing and validating the statistical analysis. GH and XJ drafted the manuscript. YS and YL reviewed and revised the manuscript critically for important intellectual content. YS and GH jointly supervised the project. All authors approved the final manuscript and agreed to be accountable for all aspects of the work.

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Data availability
No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate
This study was approved by the Ethics Committee of Wuxi Central Rehabilitation Hospital (WXMHCIRB2024LLky018). All procedures followed were in accordance with the ethical standards of the institutional research committee and the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

Competing interests
The authors declare no competing interests.

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